Ali Janati

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EDUCATION

Columbia University — Aug 2023 - Dec 2024
MSc. in Data Science — *New York, NY*

- Relevant coursework: Algorithms for Data Science, Applied Machine Learning, Applied Deep Learning, Deep Learning for NLP.
- Research Project: Working under Prof. Pierre Gentine as a Graduate Research assistant to fine-tune IBM - NASA geospatial Vision Transformer (ViT) foundation model for recognizing wind damaged forest areas through image segmentation.

Mines Paris — Sep 2018 - Jun 2023
MSc. and BSc. in Applied Maths and Computer Science — *Paris, France*

- French Top-ranking Master & Bachelor of Science, "Grande Ecole" program. Joint Master of Engineering with **ESPCI Paris.**
- Relevant coursework: Applied Mathematics, Machine Learning, Applied statistics, Computer Science, Fundamental Physics.

Lycee Lakanal — Aug 2015 - Aug 2018
BSc. in Applied Physics — *Paris, France*

- Undergraduate program preparing for the highly competitive nationwide entrance exams to the Top-ranking French Schools of Engineering. (CPGE PCSI/ PC*).
- Relevant coursework: Mathematics, Physics, Chemistry, Computer Science, Linear Algebra, Calculus.

WORK EXPERIENCE

Stealth Startup — Nov 2025 - Present
Founding Engineer, Senior Machine Learning Scientist — *San Francisco, CA*

- Fine-tuned a Dreamer-inspired world model by introducing a custom state-prediction head for smart-home state forecasting, developed custom RL environments and reward functions, and optimized policies using GRPO.
- Built an ONNX + TensorRT inference stack from scratch for on-device deployment, leveraging graph-level optimization and mixed-precision execution to reduce time-to-first-token from ~820ms to ~310ms and improve steady-state latency by ~62% compared to plain PyTorch.
- Developed a ReAct-style agentic reasoning framework with schema-constrained tool invocation, fine-tuned the language model on specific tool calling tasks, reducing invalid executions from ~12% to <1% and enabling reliable multi-step decision making.
- Designed a real-time speech interaction stack integrating VAD, ASR, and TTS, improving speech-boundary detection and streaming robustness, reducing end-to-end response time from ~1.6s to ~1.1s and eliminating clipped initial utterances.

Kaliber Labs — Feb 2025 - Nov 2025
Senior Machine Learning Scientist (Jun 2025 - Nov 2025) — *San Francisco, CA*

- Led end-to-end architecture of speech-to-speech pipelines using Distil-Whisper (ASR), LLaMA-3.2 (LLM), and CSM (TTS).
- Achieved 20% reduction in latency through quantization, speculative decoding, KV cache handling, and FlashAttention-2.
- Designed advanced TTS via emotion detection, voice identification, voice cloning, interruption handling, and always-on listening.
- Mentored 8 engineers and spearheaded delivery of production-grade pipelines for robotic interaction in noisy environments.
- Engineered and deployed a face de-identification system for Sutter Health operating rooms to detect and blur faces of surgeons, staff, and patients in surgical recordings to preserve privacy, achieving ~40 ms per-frame latency on ~30 FPS video streams.

Machine Learning Research Engineer (Feb 2025 - Jun 2025) — *San Francisco, CA*

- Built the foundational conversational AI stack of the company, from benchmarking state of the art models to deploying them.
- Drove technical roadmap and managed delivery of core features for high-stakes demos, including mentorship of junior engineers.
- Benchmarked and fine-tuned VLA models (e.g., pi-0, GR00T N1) and integrated them into simulation environments.

Esperanto Technologies — Jun 2024 - Jan 2025
Machine Learning Engineer Intern — *Mountain View, CA*

- Fine-tuned different versions of the OpenAI Whisper speech recognition model through transfer learning on domain specific datasets to enhance performance on medical conversations between doctors and patients. Achieved 14% WER drop on the validation set. Open Sourced the Model on Hugging Face and reached 35k downloads in less than 2 years.
- Identified layers that can be deleted in Llama 3.1 through several pruning techniques while keeping the highest MMLU Accuracy possible, healing the pruned model through PEFT to recover any lost performance induced by the pruning process.

Humanitics.ai, Station F — Apr 2022 - Oct 2022
Machine Learning Engineer Intern — *Paris, France*

- Contributed to Humanitics.ai, a retail-tech within world's largest startup campus: station F, and laureate of future 40 most promising startups in France in 2022.
- Shaped a traffic forecasting model to determine for each one of 500 stores, and each hour of the day in the future, the number of visitors leveraging sophisticated naïve models, prophet and PyTorch and exceeded performance expectations by 7% MAPE drop.

PUBLICATIONS

- Dumont, E. L. P., Janati, A., Bhattacharya, M., Jeannin, J.-B., Do, C. (2024). Improving allele-specific epigenomic signal coverage by 10-fold using Hidden Markov Modeling and Machine Learning. bioRxiv, https://doi.org/10.1101/2024.05.23.595536
- Grzeczkowicz, R., Soriano, E., Janati, A., Zhang, M., et al. (2026). Uncertainty-Aware Multimodal Emotion Recognition through Dirichlet Parameterization. arXiv, https://doi.org/10.48550/arXiv.2602.09121
- Janati, A., El Maghraoui, K., Luo, X., Shen, W., Zou, O., Mao, Y. (2026). Router Sensitivity Under Lightweight Fine-Tuning Identifies Prunable Experts in Mixture-of-Experts Models. arXiv, https://doi.org/10.48550/arXiv.2608.07890
- Janati, A., Kanavalau, A., El Maghraoui, K., Belfatmi, A. (2026). Post-Grokking Collapse at the Representation-Readout Interface in Muon-Trained Transformers. arXiv, https://doi.org/10.48550/arXiv.2608.07436

SKILLS

Programming: Fluent in Python (Numpy, Pandas, ScikitLearn, PyTorch, Plotly, Hugging Face), Git, Google Cloud.
Languages: English (fluent, TOEFL score: 109/120). French and Arabic (native speaker). Proficient in Spanish.
Associative Work: Tutoring for preparatory classes (CPGE) and high school students, Data for Good.
Sports: Soccer (played in Raja Casablanca at a national level from 2009 to 2014), Fitness, Martial Arts.